



**CHAPTER 9**

**FUNDAMENTALS OF TRIGONOMETRY**

### Introduction:

1. Word trigonometry is made by combining three Greek words

Trei → three

Goni → angles

Metron → measurement

Which means measurement of triangle.

2. It is an important branch of mathematics and base of calculus.
3. It is used in Business, Engineering, Surveying, Navigation, Astronomy, Physical and Social Sciences.

#### SHORT QUESTION

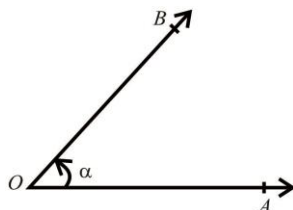
*Define trigonometry? RWP 2023, SWL 2023*

### Concept of an Angle:

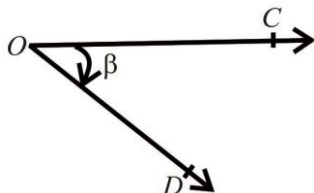
Two rays with a common starting point form an angle. One of the rays of angle is called initial side and other as terminal side.

An angle may be negative or positive due to direction of rotation.

An angle is said to be positive if its direction is anti-clock wise (counter clockwise)



An angle is said to be negative if its direction is clock wise.



Common starting point is called **vertex** of angle.

Usually angle is denoted by  $\alpha, \beta, \gamma, \theta, \phi$  etc.

### Systems of Measure of Angles:

1. Sexagesimal System.
2. Circular System.

#### SHORT QUESTION

*Define degree and radian? SGD 2022*

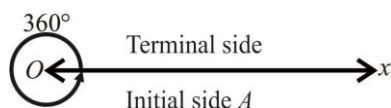
**1. Sexagesimal System** ( $D^{\circ}M'S''$ ).

In this system angle is measured in degree, minutes and seconds.

**Degree:**

If we divide the circumference of a circle into 360 equal arcs then central angle formed by each arc is called degree.

If the initial side rotates in anti-clock wise direction in such a way that it coincides with itself, the angle so formed is said to be of  $360^{\circ}$ .



So

One rotation (anti-clock wise) =  $360^{\circ}$

$\frac{1}{2}$  rotation (anti-clock wise) =  $180^{\circ}$  (Straight angle)

$\frac{1}{4}$  rotation (anti-clock wise) =  $90^{\circ}$  (Right angle)

Also 1 minutes is equal to  $60^{\text{th}}$  part of  $1^{\circ}$

$$1' = \left(\frac{1}{60}\right)^{\circ} \Rightarrow 1^{\circ} = 60'$$

1 second is equal to  $60^{\text{th}}$  part of 1 minute

$$1'' = \left(\frac{1}{60}\right)' \Rightarrow 1' = 60''$$

$$1'' = \left(\frac{1}{3600}\right)^{\circ} \Rightarrow 1^{\circ} = 3600''$$

**2. Circular System:**

In this system angle is measured in radians.

Radian is the measure of angle subtended at the centre of circle by an arc whose length is equal to its radius.

**Relation between the Length of an arc of a Circle and the Circular of its Central Angle:**

$$\ell = r\theta$$

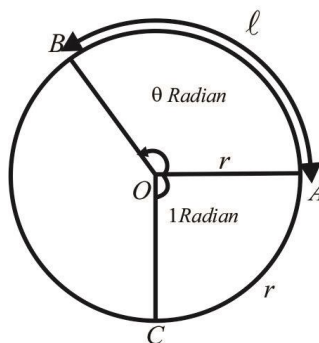
where  $\ell$  is arc length  $r$  is radius of circle  $\theta$  is angle in radians

**Proof:**

Consider a circle with centre 'O' and radius ' $r$ '.

Suppose that  $AB = \ell$  and central angle  $m\angle AOB = \theta$  radian

Take an arc AC of length =  $r$



By definition  $m\angle AOC = 1 \text{ radian}$

We know that from elementary geometry that measures of central angles of the arcs of a circle are proportional to the lengths of their arcs.

$$\frac{m\angle AOB}{m\angle AOC} = \frac{mAB}{mAC}$$

$$\frac{\theta \text{ rad}}{1 \text{ rad}} = \frac{\ell}{r}$$

$$\theta = \frac{\ell}{r}$$

$\Rightarrow$

$$\ell = r\theta$$

**Remember:**

In  $\theta = \frac{\ell}{r}$ ,  $\ell$  and  $r$  have same units so  $\theta$  is unit-less i.e. a real number.

**Relationship between Degree and Radians:**

We know that circumference of a circle with radius  $r = 2\pi r$

Let  $\ell = 2\pi r$

and angle formed by one complete revolution  $= \theta \text{ rad}$

So

$$\theta = \frac{\ell}{r}$$

$$\theta = \frac{2\pi r}{r}$$

$$\theta = 2\pi \text{ radians}$$

we can write

$$2\pi \text{ radian} = 360^\circ$$

$$\boxed{\pi \text{ rad} = 180^\circ}$$

**Deductions:**

$$(i) \quad 1 \text{ rad} = \frac{180^\circ}{\pi}$$

$$1 \text{ rad} \approx \frac{180^\circ}{3.1416}$$

$$\boxed{1 \text{ rad} \approx 57.296^\circ}$$

$$(ii) \quad 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$= \frac{3.1416}{180} \text{ rad}$$

$$\boxed{1^\circ \approx 0.01745 \text{ rad}}$$

**EXERCISE 9.1****Q.1** Explain the following measure of angles in radians**(i) 30°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 30^\circ &= 30 \times \frac{\pi}{180} \\ &= \frac{\pi}{6} \text{ rad.}\end{aligned}$$

**(ii) 45°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 45^\circ &= 45 \times \frac{\pi}{180} \\ &= \frac{\pi}{4} \text{ rad.}\end{aligned}$$

**(iii) 60°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 60^\circ &= 60 \times \frac{\pi}{180} \\ &= \frac{\pi}{3} \text{ rad.}\end{aligned}$$

**(iv) 75°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 75^\circ &= 75 \times \frac{\pi}{180} \\ &= \frac{5\pi}{12} \text{ rad.}\end{aligned}$$

**(v) 90°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 90^\circ &= 90 \times \frac{\pi}{180} \\ &= \frac{\pi}{2} \text{ rad.}\end{aligned}$$

**(vi) 105°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 105^\circ &= 105 \times \frac{\pi}{180} \\ &= \frac{7\pi}{12} \text{ rad.}\end{aligned}$$

**(vii) 120°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 120^\circ &= 120 \times \frac{\pi}{180} \\ &= \frac{2\pi}{3} \text{ rad.}\end{aligned}$$

**(viii) 135°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 135^\circ &= 135 \times \frac{\pi}{180} \\ &= \frac{3\pi}{4} \text{ rad.}\end{aligned}$$

**(ix) 150°****Solution:** As  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } 150^\circ &= 150 \times \frac{\pi}{180} \\ &= \frac{5\pi}{6} \text{ rad.}\end{aligned}$$

**(x) 10°15'**

$$\begin{aligned}\text{Solution: } 10^\circ 15' &= \left(10 + \frac{15}{60}\right)^\circ \\ &= \left(10 + \frac{1}{4}\right)^\circ \\ &= \left(\frac{40+1}{4}\right)^\circ \\ &= \left(\frac{41}{4}\right)^\circ\end{aligned}$$

**As**  $1^\circ = \frac{\pi}{180}$  radians

$$\begin{aligned}\text{So } \left(\frac{41}{4}\right)^\circ &= \frac{41}{4} \times \frac{\pi}{180} \\ &= \frac{41\pi}{720} \text{ rad.}\end{aligned}$$

**(xi)  $35^{\circ}20'$** 

$$\begin{aligned}\text{Solution: } 35^{\circ}20' &= \left(35 + \frac{20}{60}\right)^{\circ} \\ &= \left(35 + \frac{1}{3}\right)^{\circ} \\ &= \left(\frac{105+1}{3}\right)^{\circ} \\ &= \left(\frac{106}{3}\right)^{\circ}\end{aligned}$$

$$\text{As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } \left(\frac{106}{3}\right)^{\circ} &= \frac{106}{3} \times \frac{\pi}{180} \\ &= \frac{53\pi}{270} \text{ rad.}\end{aligned}$$

**(xii)  $75^{\circ}6'30''$  LHR 2022****Solution:**

$$\begin{aligned}75^{\circ}6'30'' &= \left(75 + \frac{6}{60} + \frac{30}{3600}\right)^{\circ} \\ &= \left(75 + \frac{1}{10} + \frac{1}{120}\right)^{\circ} \\ &= \left(\frac{9000+12+1}{120}\right)^{\circ} \\ &= \left(\frac{9013}{120}\right)^{\circ}\end{aligned}$$

$$\text{As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } \left(\frac{9013}{120}\right)^{\circ} &= \frac{9013}{120} \times \frac{\pi}{180} \\ &= \frac{9013\pi}{21600} \text{ rad.}\end{aligned}$$

**(xiii)  $120'40''$** **Solution:**

$$\begin{aligned}120'40'' &= \left(\frac{120}{60} + \frac{40}{3600}\right)^{\circ} \\ &= \left(2 + \frac{1}{90}\right)^{\circ} \\ &= \left(\frac{180+1}{90}\right)^{\circ}\end{aligned}$$

$$= \left(\frac{181}{90}\right)^{\circ}$$

$$\text{As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } \left(\frac{181}{90}\right)^{\circ} &= \frac{181}{90} \times \frac{\pi}{180} \\ &= \frac{181\pi}{16200} \text{ rad.}\end{aligned}$$

**(xiv)  $154^{\circ}20''$** **Solution:**

$$\begin{aligned}154^{\circ}20'' &= \left(154 + \frac{20}{3600}\right)^{\circ} \\ &= \left(154 + \frac{1}{180}\right)^{\circ} \\ &= \left(\frac{27720+1}{180}\right)^{\circ} \\ &= \left(\frac{27721}{180}\right)^{\circ}\end{aligned}$$

$$\text{As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } \left(\frac{27721}{180}\right)^{\circ} &= \frac{27721}{180} \times \frac{\pi}{180} \\ &= \frac{27721\pi}{32400} \text{ rad.}\end{aligned}$$

**(xv)  $0^{\circ}$** 

$$\text{Solution: As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } 0^{\circ} &= 0 \times \frac{\pi}{180} \text{ radian} \\ &= 0 \text{ rad.}\end{aligned}$$

**(xvi)  $3''$** 

$$\begin{aligned}\text{Solution: } 3'' &= \left(\frac{3}{3600}\right)^{\circ} \\ &= \left(\frac{1}{1200}\right)^{\circ}\end{aligned}$$

$$\text{As } 1^{\circ} = \frac{\pi}{180} \text{ radians}$$

$$\begin{aligned}\text{So } \left(\frac{1}{1200}\right)^{\circ} &= \frac{1}{1200} \times \frac{\pi}{180} \\ &= \frac{\pi}{216000} \text{ rad.}\end{aligned}$$

**Q.2** Convert the following into the measures of sexagesimal system.

(i)  $\frac{\pi}{8}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{\pi}{8} &= \frac{\pi}{8} \times \left(\frac{180}{\pi}\right)^\circ \\ &= (22.5)^\circ \\ &= 22^\circ 30'\end{aligned}$$

(ii)  $\frac{\pi}{6}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{\pi}{6} &= \frac{\pi}{6} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 30^\circ\end{aligned}$$

(iii)  $\frac{\pi}{4}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{\pi}{4} &= \frac{\pi}{4} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 45^\circ\end{aligned}$$

(iv)  $\frac{\pi}{3}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{\pi}{3} &= \frac{\pi}{3} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 60^\circ\end{aligned}$$

### SHORT QUESTION

*Convert 3 radian into degree. SWL 2023*

(v)  $\frac{\pi}{2}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{\pi}{2} &= \frac{\pi}{2} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 90^\circ\end{aligned}$$

(vi)  $\frac{2\pi}{3}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{2\pi}{3} &= \frac{2\pi}{3} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 120^\circ\end{aligned}$$

(vii)  $\frac{3\pi}{4}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{3\pi}{4} &= \frac{3\pi}{4} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 135^\circ\end{aligned}$$

(viii)  $\frac{5\pi}{6}$

**Solution:**

$$\text{As } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{5\pi}{6} &= \frac{5\pi}{6} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 150^\circ\end{aligned}$$

$$\text{(ix) } \frac{7\pi}{12}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{7\pi}{12} &= \frac{7\pi}{12} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 105^\circ\end{aligned}$$

$$\text{(x) } \frac{9\pi}{5}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{9\pi}{5} &= \frac{9\pi}{5} \times \left(\frac{180}{\pi}\right)^\circ \\ &= 324^\circ\end{aligned}$$

$$\text{(xi) } \frac{11\pi}{27}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\begin{aligned}\text{So } \frac{11\pi}{27} &= \frac{11\pi}{27} \times \left(\frac{180}{\pi}\right)^\circ \\ &= \left(\frac{220}{3}\right)^\circ \\ &= \left(73\frac{1}{3}\right)^\circ \\ &= 73^\circ \left(\frac{1}{3} \times 60\right)' \\ &= 73^\circ 20'\end{aligned}$$

$$\text{(xii) } \frac{13\pi}{16}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\text{So } \frac{13\pi}{16} = \frac{13\pi}{16} \times \left(\frac{180}{\pi}\right)^\circ$$

$$= \left(\frac{2340}{16}\right)^\circ$$

$$= \left(146\frac{1}{4}\right)^\circ$$

$$= 146^\circ \left(\frac{1}{4} \times 60\right)'$$

$$= 146^\circ 15'$$

$$\text{(xiii) } \frac{17\pi}{24}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\frac{17\pi}{24} = \frac{17\pi}{24} \times \left(\frac{180}{\pi}\right)^\circ$$

$$= \left(\frac{255}{2}\right)^\circ$$

$$= \left(127\frac{1}{2}\right)^\circ$$

$$= 127^\circ \left(\frac{1}{2} \times 60\right)'$$

$$= 127^\circ 30'$$

$$\text{(xiv) } \frac{25\pi}{36}$$

**Solution:**

$$\text{As 1 radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$\text{So } \frac{25\pi}{36} = \frac{25\pi}{36} \times \left(\frac{180}{\pi}\right)^\circ$$

$$= 125^\circ$$

(xv)  $\frac{19\pi}{32}$

**Solution:**

As 1 radian =  $\left(\frac{180}{\pi}\right)^\circ$

So  $\frac{19\pi}{32} = \frac{19\pi}{32} \times \left(\frac{180}{\pi}\right)^\circ$

$= \left(106\frac{7}{8}\right)^\circ$

$= 106^\circ \left(\frac{7}{8} \times 60\right)'$

$= 106^\circ \left(\frac{105}{2}\right)'$

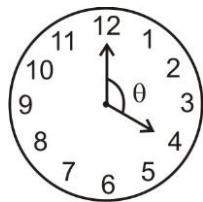
$= 106^\circ \left(52\frac{1}{2}\right)'$

$= 106^\circ 52' \left(\frac{1}{2} \times 60\right)''$

$= 106^\circ 52' 30''$

**Q.3** What is the circular measure of the angle between the hands of a clock at 4 O' clock? *SGD 2021*

**Solution:**



By fixing minute hand at 12

Angle measured in 12 hours =  $2\pi$  rad

Angle measured in 1 hour

$= \frac{2\pi}{12} = \frac{\pi}{6}$

Angle measured at 4 O' clock

$= 4 \times \frac{\pi}{6} = \frac{2\pi}{3}$

**Q.4** Find  $\theta$  when

(i)  $l = 1.5\text{cm}, r = 2.5\text{cm}$

*FSD 2021, RWP 2023*

**Solution:**

$l = 1.5\text{cm}, r = 2.5\text{cm}$

Using  $\theta = \frac{l}{r}$

$= \frac{1.5}{2.5}$

$= 0.6 \text{ radians.}$

(ii)  $l = 3.2\text{m}, r = 2\text{m}$  *FSD 2022*

**Solution:**

$l = 3.2\text{m}, r = 2\text{m}$

Using  $\theta = \frac{l}{r}$

$= \frac{3.2}{2}$

$= 1.6 \text{ rad.}$

**Q.5** Find  $l$  when

(i)  $\theta = \pi, r = 6\text{cm}$

**Solution:**

$\theta = \pi, r = 6\text{cm}$

Using  $\theta = \frac{l}{r}$

$l = r\theta$

$= 6\pi$

$= 6(3.1416)$

$= 18.86\text{cm}$

(ii)  $\theta = 65^\circ 20', r = 18\text{mm}$

*GRW 2022, BWP 2022, FSD 2023*

**Solution:**

$r = 18\text{mm}$

$\theta = 65^\circ 20'$

$= \left(65 + \frac{20}{60}\right)^\circ$

$= \left(65 + \frac{1}{3}\right)^\circ$



$$= \left( \frac{195+1}{3} \right)^\circ$$

$$= \left( \frac{196}{3} \right)^\circ$$

$$= \frac{196}{3} \times \frac{\pi}{180}$$

$$= \frac{49\pi}{135}$$

Using  $\theta = \frac{\ell}{r}$

$$\ell = r\theta$$

$$= (18) \left( \frac{49\pi}{135} \right)$$

$$\ell \approx 20.52 \text{ mm}$$

**Q.6 Find  $r$  when**

(i)  $l = 5 \text{ cm}, \theta = \frac{1}{2} \text{ radian}$

*LHR 2022, MTN 2022*

**Solution:**

$$\ell = 5 \text{ cm}, \theta = \frac{1}{2} \text{ radian}$$

Using  $\theta = \frac{\ell}{r}$

$$r = \frac{l}{\theta}$$

$$r = \frac{5}{\frac{1}{2}}$$

$$= 10 \text{ cm}$$

(ii)  $l = 56 \text{ cm}, \theta = 45^\circ$

*GRW 2022, MTN 2022, RWP 2022*

**Solution:**

$$\ell = 56 \text{ cm}$$

$$\theta = 45^\circ$$

$$= 45 \times \frac{\pi}{180}$$

$$= \frac{\pi}{4}$$

Using  $\theta = \frac{\ell}{r}$

$$r = \frac{\ell}{\theta}$$

$$= \frac{56}{\frac{\pi}{4}}$$

$$= \frac{224}{\pi}$$

$$r = 71.30 \text{ cm}$$

**Q.7 What is the length of the arc intercepted on a circle of radius 14 cm by the arms of central angle of  $45^\circ$ . *GRW 2023, MTN 2023***

**Solution:**

$$r = 14 \text{ cm}, \ell = ?$$

$$\theta = 45^\circ$$

$$= 45 \times \frac{\pi}{180}$$

$$= \frac{\pi}{4}$$

Using  $\theta = \frac{\ell}{r}$

$$\ell = r\theta$$

$$= 14 \times \frac{\pi}{4}$$

$$\ell \approx 11 \text{ cm}$$

**Q.8 Find the radius of the circle, in which the arms of a central angle of measure 1 radian cut off an arc of length 35 cm. *DGK 2022***

**Solution:**

$$\ell = 35 \text{ cm}, r = ?$$

$$\theta = 1 \text{ rad}$$

Using

$$\theta = \frac{\ell}{r}$$

$$r = \frac{\ell}{\theta}$$

$$r = \frac{35}{1}$$

$$r = 35\text{cm}$$

**Q.9** A railway train is running on a circular track of radius 500m at the rate of 30km/h. through what angle will it turn in 10 sec?

RWP 2023, BWP 2023

**Solution:**

$$r = 500\text{m}$$

$$\text{speed of train} = 30\text{kmh}^{-1}$$

$$= \frac{30 \times 1000}{3600} \text{ms}^{-1}$$

$$= \frac{25}{3} \text{ms}^{-1}$$

and

$$\ell = s = r\theta$$

$$\ell = \frac{25}{3} \times 10$$

$$= \frac{250}{3} \text{m}$$

Using

$$\theta = \frac{\ell}{r}$$

$$\theta = \frac{\frac{250}{3}}{500}$$

$$= \frac{1}{6} \text{rad}$$

**Q.10** A horse is tethered to a peg by a rope of 9m length and it can move in a circle with the peg as centre. If the horse moves along the circumference of the circle,

keeping the rope tight how far will it have gone when the rope has turned through angle of  $70^\circ$ ?

**Solution:**

Here

$$\theta = 70^\circ$$

$$= 70 \times \frac{\pi}{180}$$

$$= \frac{7\pi}{18}$$

And

$$r = 9\text{m}$$

Using

$$\theta = \frac{\ell}{r}$$

$$\ell = r\theta$$

$$= (9) \left( \frac{7\pi}{18} \right)$$

$$= \frac{7\pi}{2}$$

$$\ell \approx 11\text{m}$$

**Q.11** The pendulum of a clock is 20cm long and it swings through an angle of  $20^\circ$  each second. How far does tip of the pendulum move in 1 sec?

SWL 2022

**Solution:**

Here

$$\theta = 20^\circ$$

$$= 20 \times \frac{\pi}{180}$$

$$= \frac{\pi}{9}$$

And

$$r = 20\text{cm}$$

Using

$$\theta = \frac{\ell}{r}$$

$$\ell = r\theta$$

$$= (20) \left( \frac{\pi}{9} \right)$$

$$\ell \approx 6.98 \text{ cm}$$

**Q.12** Assuming the average distance of the earth from the sun to be  $148 \times 10^6$  km and the angle subtended by the sun at the eye of a person on the earth is of measure  $9.3 \times 10^{-3}$  radians. Find the diameter of the sun.

**Solution:**

Let O be the eye of observer on earth,  $r$  be the distance between earth and sun.  $\theta$  is the angle subtended by the sun on the earth.



Here

$$r = 148 \times 10^6 \text{ km}$$

$$\theta = 9.3 \times 10^{-3} \text{ rad}$$

Using

$$\theta = \frac{\ell}{r}$$

So

$$\ell = r\theta$$

$$\ell = (148 \times 10^6)(9.3 \times 10^{-3})$$

$$= 1.37 \times 10^6 \text{ km}$$

But here  $\ell \approx d$  (diameter of the sun)

**Q.13** A circular wire of radius 6cm is cut straightened and then bent so as to lie along the circumference of a hoop of radius 24cm. Find the measure of the angle which it subtends at the centre of the hoop.

**Solution:**

Here

$$r = 6$$

Circumference of hoop:

$$C = \ell = 2\pi r = 2\pi(6) = 12\pi$$

Radius of hoop

$$r = 24 \text{ cm}$$

Using

$$\theta = \frac{\ell}{r}$$

$$= \frac{12\pi}{24}$$

$$\theta = \frac{\pi}{2}$$

**Q.14** Show that the area of a sector of a circular region of radius  $r$  is  $\frac{1}{2}r^2\theta$ , where  $\theta$  is the circular measure of the central angle of sector. DGK 2023

**Solution:**

From elementary geometry, we know that

$$\frac{\text{area of sector}}{\text{area of circle}} = \frac{\text{angle subtended by sector}}{\text{angle subtended by circle}}$$

$$\frac{\text{area of sector}}{\pi r^2} = \frac{\theta}{2\pi}$$

$$\text{area of sector} = \frac{1}{2} r^2 \theta$$

- Q.15** Two cities A and B lie on the equator such that their longitudes are  $45^\circ\text{E}$  and  $25^\circ\text{W}$  respectively. Find the distance between the two cities, taking radius of earth as 6400km. LHR 2021

**Solution:**

Here Radius of earth;  $r = 6400\text{Km}$

Angel between city A and B :  $\theta = 45^\circ + 25^\circ$

$$\theta = 70^\circ$$

$$= 70 \times \frac{\pi}{180} \text{ rad}$$

$$= \frac{7\pi}{18} \text{ rad}$$

Let  $\ell$  be the distance between city A and B

Using  $\theta = \frac{\ell}{r}$

$$\ell = r\theta$$

$$= 6400 \times \frac{7\pi}{18}$$

$$\ell \approx 7819\text{km}$$

But  $\ell$  is an approximate distance between two cities, so  $|AB| \approx 7819\text{km}$

- Q.16** The moon subtends an angle of  $0.5^\circ$  at the eye of an observer on earth. The distance of the moon from earth is  $3.844 \times 10^5\text{km}$ . What is the length of the diameter of the moon?

**Solution:**



Let O be the eye of observer on earth, r be the distance between earth and moon.

$\theta$  be the angle subtended by moon on the earth.

Here

$$\theta = 0.5^\circ$$

$$= \frac{1}{2} \times \frac{\pi}{180}$$

$$\theta = \frac{\pi}{360}$$

Using

$$\theta = \frac{\ell}{r}$$

$$\ell = r\theta$$

$$= 3.844 \times 10^5 \times \frac{\pi}{360}$$

$$\ell \approx 3354.31 \text{ km}$$

**Q.17** The angle subtended by the earth at the eye of a space man, landed on the moon is  $1^\circ 54'$ . The radius of the earth is 6400 km. Find the approximate distance between the moon and earth.

**Solution:**

Let O be the eye of observer on moon,  $\theta$  be the angle subtended by earth on moon.

$l$  be the diameter of earth and  $r$  be the distance between moon and earth.

Here  $\ell$  = diameter of earth  $= 2 \times 6400 = 12800 \text{ km}$



$$\theta = \left(1 + \frac{54}{60}\right)^\circ$$

$$= \left(1 + \frac{9}{10}\right)^\circ$$

$$= \left(\frac{19}{10}\right)^\circ$$

$$\theta = \frac{19}{10} \times \frac{\pi}{180} \text{ rad}$$

$$= \frac{19\pi}{1800}$$

Using

$$\theta = \frac{\ell}{r}$$

$$r = \frac{\ell}{\theta}$$

$$r \approx \frac{12800}{\frac{19\pi}{1800}}$$

$$r \approx 385993 \text{ km}$$

So the distance between earth and moon is approximately  $3.85993 \times 10^5$  km.

**General Angle (Co-terminal Angles):**

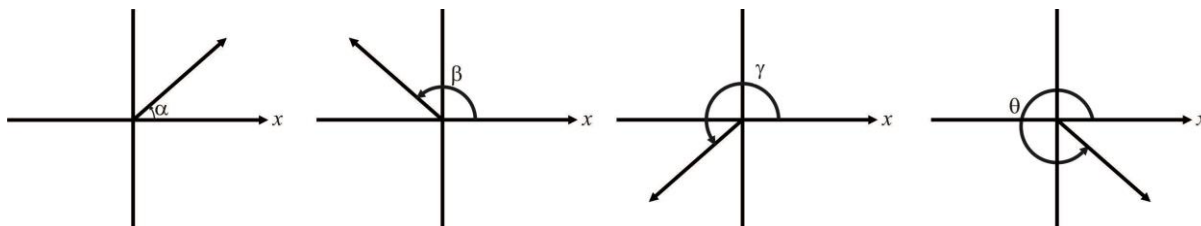
Angles with the same initial and terminal sides are called co-terminal angles.

General angle is  $\theta + 2k\pi, k \in \mathbb{Z}$ . Where  $0 \leq \theta \leq 2\pi$

**Angle in Standard Position:**

An angle is said to be in standard position if its vertex lies at the origin of a rectangular coordinate system and its initial side along positive  $x$ -axis.

An angle in standard position is said to lie in a quadrant due to its terminal side as



Angle  $\alpha$  lies in I Quadrant as its terminal side lies in I Quadrant

Angle  $\beta$  lies in II Quadrant as its terminal side lies in II Quadrant

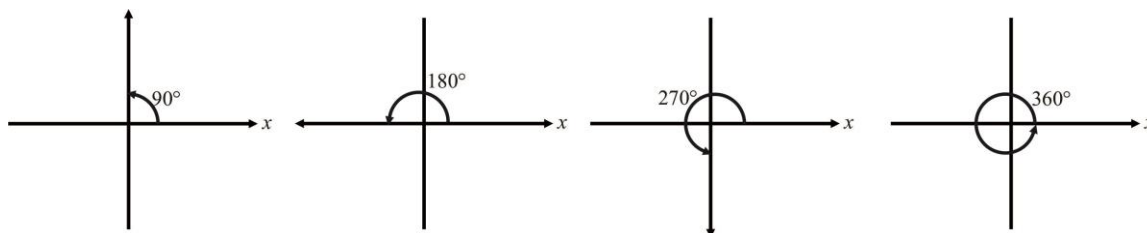
Angle  $\gamma$  lies in III Quadrant as its terminal side lies in III Quadrant

Angle  $\theta$  lies in IV Quadrant as its terminal side lies in IV Quadrant

**Quadrantal angle:**

An angle in standard position is said to be quadrantal angle if its terminal side lies along  $x$ -axis or  $y$ -axis.

Such as  $0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ, 450^\circ, \dots$

**Trigonometric Functions:**

If  $m\angle BAC = \theta$  is fixed angle in right angle triangle  $ABC$

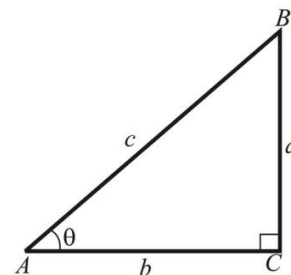
And  $m\overline{AB} = c$   $m\overline{BC} = a$   $m\overline{AC} = b$

Six ratios can be formed by using all three

sides of a triangle as  $\frac{a}{b}, \frac{a}{c}, \frac{b}{c}, \frac{b}{a}, \frac{c}{a}, \frac{c}{b}$

These ratios are called **trigonometric functions** of angle  $\theta$  and are defined as below

$$\sin \theta = \frac{a}{c} = \frac{\text{opp}}{\text{hyp}}, \quad \text{cosec } \theta = \frac{c}{a} = \frac{\text{hyp}}{\text{opp}}$$



$$\cos \theta = \frac{b}{c} = \frac{\text{adj}}{\text{hyp}}, \quad \sec \theta = \frac{c}{b} = \frac{\text{hyp}}{\text{adj}}$$

$$\tan \theta = \frac{a}{b} = \frac{\text{opp}}{\text{adj}}, \quad \cot \theta = \frac{b}{a} = \frac{\text{adj}}{\text{opp}}$$

We also have relations between these six trigonometric function as follows:

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad ; \quad \sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$\sec \theta = \frac{1}{\cos \theta} \quad ; \quad \cos \theta = \frac{1}{\sec \theta}$$

$$\cot \theta = \frac{1}{\tan \theta} \quad ; \quad \tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta} \quad ; \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$

### **Fundamental Identities:**

(1)  $\sin^2 \theta + \cos^2 \theta = 1$

(2)  $1 + \tan^2 \theta = \sec^2 \theta$

(3)  $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

### **Proof:**

(1) Refer to right angle triangle  $ABC$

by Pythagoras theorem  $a^2 + b^2 = c^2$

(i)

divide by " $c^2$ " both sides of equation (i)

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1$$

$$\text{Hence } \sin^2 \theta + \cos^2 \theta = 1$$

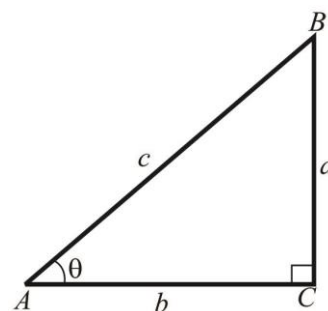
(2) Dividing by ' $b^2$ ' both sides of equation (i)

$$\frac{a^2}{b^2} + \frac{b^2}{b^2} = \frac{c^2}{b^2}$$

$$\left(\frac{a}{b}\right)^2 + 1 = \left(\frac{c}{b}\right)^2$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\text{Hence } 1 + \tan^2 \theta = \sec^2 \theta$$



(3) Dividing by ' $a^2$ ' both sides of equation (i)

$$\frac{a^2}{a^2} + \frac{b^2}{a^2} = \frac{c^2}{a^2}$$

$$1 + \left(\frac{b}{a}\right)^2 = \left(\frac{c}{a}\right)^2$$

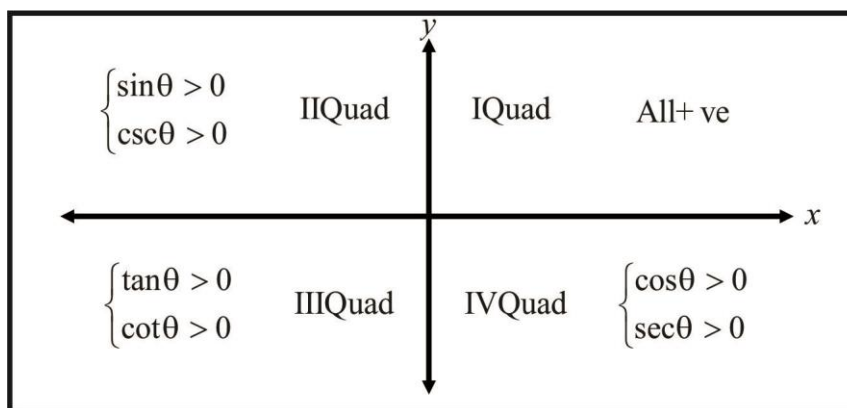
$$\text{Hence } 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

**Note:**

$$(\sin \theta)^2 = \sin^2 \theta, (\cos \theta)^2 = \cos^2 \theta \text{ and } (\tan \theta)^2 = \tan^2 \theta \text{ etc.}$$

**Signs of Trigonometric Functions:**

It  $\theta$  is in standard position and not a quadrantal angle then it will lie in a quadrant and sign of trigonometric functions in different quadrants are shown in diagram



**Results:**

$$\begin{array}{ll} \sin(-\theta) = -\sin \theta & ; \quad \operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta \\ \cos(-\theta) = \cos \theta & ; \quad \sec(-\theta) = \sec \theta \\ \tan(-\theta) = -\tan \theta & ; \quad \cot(-\theta) = -\cot \theta \end{array}$$



**EXERCISE 9.2****Q.1 Find the signs of the following.**

**(i)  $\sin 160^\circ$**

**Solution:** Positive $(\because 160^\circ \text{ lies in IIrd quadrant})$ 

**(ii)  $\cos 190^\circ$**

**Solution:** Negative $(\because 190^\circ \text{ lies in IIIrd quadrant})$ 

**(iii)  $\tan 115^\circ$**

**Solution:** Negative $(\because 115^\circ \text{ lies in IIrd quadrant})$ 

**(iv)  $\sec 245^\circ$**

**Solution:** Negative $(\because 245^\circ \text{ lies in IIIrd quadrant})$ 

**(v)  $\cot 80^\circ$**

**Solution:** Positive $(\because 80^\circ \text{ lies in Ist quadrant})$ 

**(vi)  $\operatorname{cosec} 297^\circ$**

**Solution:** Negative $(\because 297^\circ \text{ lies in IVth quadrant})$ **Q.2 Fill up the blanks in the following**

**(i)  $\sin(-310^\circ) =$**

**Solution:**  $\sin(-310^\circ) = -\sin 310^\circ$ 

**(ii)  $\tan(-182^\circ) =$**

**Solution:**  $\tan(-182^\circ) = -\tan 182^\circ$ 

**(iii)  $\cos(-75^\circ) =$**

**Solution:**  $\cos(-75^\circ) = +\cos 75^\circ$ 

**(iv)  $\cot(-137^\circ) =$**

**Solution:**  $\cot(-137^\circ) = -\cot 137^\circ$ 

**(v)  $\sec(-216^\circ) =$**

**Solution:**  $\sec(-216^\circ) = +\sec 216^\circ$ 

**(vi)  $\operatorname{cosec}(-15^\circ) =$**

**Solution:**  $\operatorname{cosec}(-15^\circ) = -\operatorname{cosec} 15^\circ$ **Q.3 In which quadrant are the terminal arms of the angle lie when.**

**(i)  $\sin \theta < 0$  and  $\cos \theta > 0$**

**Solution:**As  $\sin \theta < 0 \Rightarrow \theta$  lies in quad III or IV $\cos \theta > 0 \Rightarrow \theta$  lies in quad I or IVHence  $\theta$  lies in quad IV

**(ii)  $\cot \theta > 0$  and  $\operatorname{cosec} \theta > 0$**

**Solution:**As  $\cot \theta > 0 \Rightarrow \theta$  lies in quad I or III $\operatorname{cosec} \theta > 0 \Rightarrow \theta$  lies in quad I or IIHence  $\theta$  lies in quad I

**(iii)  $\tan \theta < 0$  and  $\cos \theta > 0$**

**Solution:**As  $\tan \theta < 0 \Rightarrow \theta$  lies in quad II or IV $\cos \theta > 0 \Rightarrow \theta$  lies in quad I or IVHence  $\theta$  lies in quad IV

**(iv)  $\sec \theta < 0$  and  $\sin \theta < 0$**

**Solution:**As  $\sec \theta < 0 \Rightarrow \theta$  lies in quad II or III $\sin \theta < 0 \Rightarrow \theta$  lies in quad III or IVHence  $\theta$  lies in quad III

**(v)  $\cot \theta > 0$  and  $\sin \theta < 0$**

**Solution:**As  $\cot \theta > 0 \Rightarrow \theta$  lies in quad I or III $\sin \theta < 0 \Rightarrow \theta$  lies in quad III or IVHence  $\theta$  lies in quad III

**(vi)  $\cos \theta < 0$  and  $\tan \theta < 0$**

**Solution:**As  $\cos \theta < 0 \Rightarrow \theta$  lies in quad II or III $\tan \theta < 0 \Rightarrow \theta$  lies in quad II or IVHence  $\theta$  lies in quad II

**Q.4** Find the values of the remaining trigonometric functions if,

(i)  $\sin \theta = \frac{12}{13}$  and terminal arm

of  $\theta$  is in 1<sup>st</sup> quadrant.

**Solution:**

Given that  $\sin \theta = \frac{12}{13}$  so

$$\operatorname{cosec} \theta = \frac{13}{12}$$

Now  $\cos^2 \theta = 1 - \sin^2 \theta$

$$\begin{aligned} &= 1 - \left(\frac{12}{13}\right)^2 \\ &= 1 - \frac{144}{169} \\ &= \frac{169 - 144}{169} \end{aligned}$$

$$\cos^2 \theta = \frac{25}{169}$$

$$\cos \theta = \pm \frac{5}{13}$$

But  $\theta$  lies in 1<sup>st</sup> quadrant, so

$$\cos \theta = \frac{5}{13} \text{ and } \sec \theta = \frac{13}{5}$$

Now  $\tan \theta = \frac{\sin \theta}{\cos \theta}$

$$\tan \theta = \frac{\frac{12}{13}}{\frac{5}{13}}$$

$$\tan \theta = \frac{12}{5} \text{ and } \cot \theta = \frac{5}{12}$$

(ii)  $\cos \theta = \frac{9}{41}$  and terminal

arm of the angle  $\theta$  is in 4<sup>th</sup> quadrant

**Solution:**

$$\cos \theta = \frac{9}{41} \text{ and } \sec \theta = \frac{41}{9}$$

Now  $\sin^2 \theta = 1 - \cos^2 \theta$

$$\begin{aligned} &= 1 - \left(\frac{9}{41}\right)^2 \\ &= 1 - \frac{81}{1681} \\ &= \frac{1681 - 81}{1681} \end{aligned}$$

$$\sin^2 \theta = \frac{1600}{1681}$$

$$\sin \theta = \pm \frac{40}{41}$$

But  $\theta$  lies in 4th quadrant,

$$\sin \theta = \frac{-40}{41} \text{ and } \operatorname{cosec} \theta = \frac{-41}{40}$$

Now  $\tan \theta = \frac{\sin \theta}{\cos \theta}$

$$\begin{aligned} &= \frac{-\frac{40}{41}}{\frac{9}{41}} \end{aligned}$$

$$\tan \theta = \frac{-40}{9} \text{ and } \cot \theta = \frac{-9}{40}$$

**Alternate Solution:**

$$\cos \theta = \frac{9}{41}, \sec \theta = \frac{41}{9}$$

Consider a right triangle,  
by using Pythagoras theorem

$$x^2 + 9^2 = 41^2$$

$$x^2 = 1681 - 81$$

$$x^2 = 1600$$

$$x = 40$$

Since  $\theta$  lies in 4<sup>th</sup> quadrant,

$$\sin \theta = -\frac{40}{41}, \operatorname{cosec} \theta = -\frac{41}{40}$$

$$\tan \theta = -\frac{40}{9}, \cot \theta = -\frac{9}{40}$$

(iii)  $\cos \theta = -\frac{\sqrt{3}}{2}$  and the  
terminal arm of the angle is  
in quad. III

**Solution:**

$$\text{Given that } \cos \theta = -\frac{\sqrt{3}}{2} \text{ so}$$

$$\sec \theta = -\frac{2}{\sqrt{3}}$$

$$\text{Now } \sin^2 \theta = 1 - \cos^2 \theta$$

$$= 1 - \left( \frac{-\sqrt{3}}{2} \right)^2$$

$$= 1 - \frac{3}{4} = \frac{4-3}{4}$$

$$\sin^2 \theta = \frac{1}{4}$$

$$\sin \theta = \pm \frac{1}{2}$$

But  $\theta$  lies in 3<sup>rd</sup> quadrant so

$$\sin \theta = \frac{-1}{2} \text{ and } \operatorname{cosec} \theta = -2$$

$$\text{Now } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}}$$

$$\tan \theta = \frac{1}{\sqrt{3}} \text{ and } \cot \theta = \sqrt{3}$$

**Alternate Solution:**

$$\cos \theta = -\frac{\sqrt{3}}{2}, \sec \theta = -\frac{2}{\sqrt{3}}$$

Consider a right triangle

Using Pythagoras theorem

$$x^2 + (\sqrt{3})^2 = (2)^2$$

$$x^2 + 3 = 4$$

$$x^2 = 4 - 3 = 1$$

$$x = 1$$

Since  $\theta$  lies in 3<sup>rd</sup> quadrant

So

$$\sin \theta = -\frac{1}{2}, \operatorname{cosec} \theta = -2$$

$$\tan \theta = \frac{1}{\sqrt{3}}, \cot \theta = \sqrt{3}$$

(iv)  $\tan \theta = \frac{-1}{3}$  and the terminal  
arm of the angle is in quad.  
II

**Solution:**

Given that

$$\tan \theta = \frac{-1}{3} \quad \text{so} \quad \cot \theta = -3$$

$$\text{Now } \sec^2 \theta = 1 + \tan^2 \theta$$

$$= 1 + \left( \frac{-1}{3} \right)^2$$

$$= 1 + \frac{1}{9}$$

$$\sec^2 \theta = \frac{10}{9}$$

$$\sec \theta = \pm \frac{\sqrt{10}}{3}$$

But  $\theta$  lies in 2<sup>nd</sup> quadrant, so

$$\sec \theta = \frac{-\sqrt{10}}{3} \text{ and } \cos \theta = \frac{-3}{\sqrt{10}}$$

$$\begin{aligned}\text{Now } \tan \theta &= \frac{\sin \theta}{\cos \theta} \\ \sin \theta &= \tan \theta \cdot \cos \theta \\ \sin \theta &= \left(-\frac{1}{3}\right) \left(\frac{-3}{\sqrt{10}}\right) \\ \sin \theta &= \frac{1}{\sqrt{10}}\end{aligned}$$

$$\text{and } \operatorname{cosec} \theta = \sqrt{10}$$

(v)  $\sin \theta = \frac{-1}{\sqrt{2}}$  and terminal arm of  $\theta$  is not in 3<sup>rd</sup> quadrant.

**Solution:**

$$\text{Given that } \sin \theta = \frac{-1}{\sqrt{2}} \text{ and}$$

$$\operatorname{cosec} \theta = -\sqrt{2}$$

$$\begin{aligned}\text{Now } \cos^2 \theta &= 1 - \sin^2 \theta \\ &= 1 - \left(\frac{-1}{\sqrt{2}}\right)^2 \\ &= 1 - \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\cos^2 \theta &= \frac{1}{2} \\ \cos \theta &= \pm \frac{1}{\sqrt{2}}\end{aligned}$$

As terminal arm of  $\theta$  is not in quad-III

So  $\theta$  lies in 4<sup>th</sup> quadrant so

$$\cos \theta = \frac{1}{\sqrt{2}} \text{ and } \sec \theta = \sqrt{2}$$

$$\text{Now } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$\tan \theta = -1 \text{ and } \cot \theta = -1$$

**Q.5** If  $\cot \theta = \frac{15}{8}$  and the terminal arm of the angle is not in quadrant I, find the values of  $\cos \theta$  and  $\operatorname{cosec} \theta$ .

**Solution:**

$$\text{Give that } \cot \theta = \frac{15}{8} \text{ so } \tan \theta = \frac{8}{15}$$

$$\text{Now } \sec^2 \theta = 1 + \tan^2 \theta$$

$$= 1 + \left(\frac{8}{15}\right)^2$$

$$= 1 + \frac{64}{225}$$

$$= \frac{225 + 64}{225}$$

$$\sec^2 \theta = \frac{289}{225}$$

$$\sec \theta = \pm \frac{17}{15}$$

As terminal arm of  $\theta$  is not in quad.

I

So  $\theta$  lies in 3<sup>rd</sup> quadrant, so

$$\sec \theta = -\frac{17}{15} \text{ and } \cos \theta = \frac{-15}{17}$$

$$\text{and } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\begin{aligned}\sin \theta &= \tan \theta \cdot \cos \theta \\ &= \left(\frac{8}{15}\right) \left(\frac{-15}{17}\right)\end{aligned}$$

$$\sin \theta = -\frac{8}{17}$$

$$\text{So } \operatorname{cosec} \theta = \frac{-17}{8}$$

**Q.6** If  $\operatorname{cosec} \theta = \frac{m^2+1}{2m}$  and  $m > 0, \left(0 < \theta < \frac{\pi}{2}\right)$ , find the values of remaining trigonometric ratios.

**Solution:**

Given that  $\operatorname{cosec} \theta = \frac{m^2+1}{2m}$  so

$$\sin \theta = \frac{2m}{1+m^2}$$

Now  $\cos^2 \theta = 1 - \sin^2 \theta$

$$\begin{aligned} &= 1 - \left(\frac{2m}{1+m^2}\right)^2 \\ &= 1 - \frac{4m^2}{1+m^4+2m^2} \\ &= \frac{1+m^4+2m^2-4m^2}{1+m^4+2m^2} \end{aligned}$$

$$\cos^2 \theta = \frac{1+m^4-2m^2}{1+m^4+2m^2}$$

$$\cos^2 \theta = \frac{(1-m^2)^2}{(1+m^2)^2}$$

$$\cos \theta = \pm \frac{1-m^2}{1+m^2}$$

But  $\theta$  lies in 1<sup>st</sup> quadrant, so

$$\cos \theta = \frac{1-m^2}{1+m^2} \text{ and } \sec \theta = \frac{1+m^2}{1-m^2}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\begin{aligned} &= \frac{\frac{2m}{1+m^2}}{\frac{1-m^2}{1+m^2}} \\ &= \frac{2m}{1-m^2} \end{aligned}$$

$$\tan \theta = \frac{2m}{1-m^2} \text{ and } \cot \theta = \frac{1-m^2}{2m}$$

**Q.7** If  $\tan \theta = \frac{1}{\sqrt{7}}$  and the terminal arm of  $\theta$  is not in the 3<sup>rd</sup> quadrant, find the value of  $\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta}$

**Solution:**

Given that  $\tan \theta = \frac{1}{\sqrt{7}}$

and  $\theta$  does not lie in 3<sup>rd</sup> quadrant, then it must lie in 1<sup>st</sup> quadrant.

Now  $\sec^2 \theta = 1 + \tan^2 \theta$

$$= 1 + \left(\frac{1}{\sqrt{7}}\right)^2$$

$$= 1 + \frac{1}{7}$$

$$\sec^2 \theta = \frac{8}{7}$$

Also,  $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

$$\begin{aligned} &= 1 + \left(\sqrt{7}\right)^2 \\ &= 1 + 7 \end{aligned}$$

$$\operatorname{cosec}^2 \theta = 8$$

$$\text{Now } \frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta} = \frac{8 - \frac{8}{7}}{8 + \frac{8}{7}}$$

$$\begin{aligned} &= \frac{56 - 8}{56 + 8} \\ &= \frac{48}{64} \end{aligned}$$

$$\begin{aligned} &= \frac{48}{64} \\ &= \frac{3}{4} \end{aligned}$$

$$\begin{aligned} &= \frac{48}{64} \\ &= \frac{3}{4} \end{aligned}$$

**Alternate Solution:**

$$\tan \theta = \frac{1}{\sqrt{7}}, \cot \theta = \sqrt{7}$$

As  $\theta$  does not lie in 3<sup>rd</sup> quadrant, so it must lie in 1<sup>st</sup> quadrant.

$$\begin{aligned} &\text{Take } \frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta} \\ &= \frac{(1 + \cot^2 \theta) - (1 + \tan^2 \theta)}{(1 + \cot^2 \theta) + (1 + \tan^2 \theta)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{7 - \frac{1}{7}}{2 + 7 + \frac{1}{7}} \\
 &= \frac{\frac{49-1}{7}}{9 + \frac{1}{7}} \\
 &= \frac{\frac{48}{7}}{\frac{64}{7}} \\
 &= \frac{48}{64} \\
 &= \frac{3}{4}
 \end{aligned}$$

**Q.8** If  $\cot \theta = \frac{5}{2}$  and the terminal arm of  $\theta$  is in the first quadrant, find the value of  $\frac{3\sin \theta + 4\cos \theta}{\cos \theta - \sin \theta}$

**Solution:**

Given that  $\cot \theta = \frac{5}{2}$  so  $\tan \theta = \frac{2}{5}$

Now  $\sec^2 \theta = 1 + \tan^2 \theta$

$$= 1 + \left(\frac{2}{5}\right)^2$$

$$= 1 + \frac{4}{25}$$

$$= \frac{25+4}{25}$$

$$\sec^2 \theta = \frac{29}{25}$$

As  $\theta$  lies in 1<sup>st</sup> quadrant

$$\text{So } \sec \theta = \frac{\sqrt{29}}{5} \text{ and } \cos \theta = \frac{5}{\sqrt{29}}$$

$$\text{Now } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin \theta = \tan \theta \cdot \cos \theta$$

$$= \left(\frac{2}{5}\right) \left(\frac{5}{\sqrt{29}}\right)$$

$$\sin \theta = \frac{2}{\sqrt{29}}$$

So

$$\frac{3\sin \theta + 4\cos \theta}{\cos \theta - \sin \theta} = \frac{3\left(\frac{2}{\sqrt{29}}\right) + 4\left(\frac{5}{\sqrt{29}}\right)}{\frac{5}{\sqrt{29}} - \frac{2}{\sqrt{29}}}$$

$$= \frac{6+20}{\frac{5-2}{\sqrt{29}}}$$

$$= \frac{26}{\frac{3}{\sqrt{29}}}$$

$$= \frac{26}{3} \cdot \frac{\sqrt{29}}{\sqrt{29}}$$

$$= \frac{26}{3} \cdot \frac{\sqrt{29}}{\sqrt{29}}$$

**Alternate Solution:**

$$\cot \theta = \frac{5}{2}$$

**Take**  $\frac{3\sin \theta + 4\cos \theta}{\cos \theta - \sin \theta}$

**Dividing up and down by  $\sin \theta$**

$$\frac{3\sin \theta + 4\cos \theta}{\cos \theta - \sin \theta} = \frac{\sin \theta}{\frac{\cos \theta - \sin \theta}{\sin \theta}}$$

$$= \frac{\sin \theta}{\frac{3 + 4\cot \theta}{\cot \theta - 1}}$$

$$= \frac{3 + 4\left(\frac{5}{2}\right)}{\frac{5}{2} - 1}$$

$$= \frac{6+20}{\frac{5-2}{2}}$$

$$= \frac{26}{\frac{3}{2}}$$

$$= \frac{26}{3} \cdot \frac{2}{2}$$

The values of trigonometric functions of acute angles:

| $\theta$                      | $30^\circ$           | $45^\circ$           | $60^\circ$           |
|-------------------------------|----------------------|----------------------|----------------------|
| $\sin \theta$                 | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ |
| $\cos \theta$                 | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        |
| $\tan \theta$                 | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           |
| $\cot \theta$                 | $\sqrt{3}$           | 1                    | $\frac{1}{\sqrt{3}}$ |
| $\sec \theta$                 | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$           | 2                    |
| $\operatorname{cosec} \theta$ | 2                    | $\sqrt{2}$           | $\frac{2}{\sqrt{3}}$ |

The values of trigonometric function of angles  $0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$ :

| $\theta$                      | $0^\circ$ | $90^\circ$ | $180^\circ$ | $270^\circ$ | $360^\circ$ |
|-------------------------------|-----------|------------|-------------|-------------|-------------|
| $\sin \theta$                 | 0         | 1          | 0           | -1          | 0           |
| $\cos \theta$                 | 1         | 0          | -1          | 0           | 1           |
| $\tan \theta$                 | 0         | $\infty$   | 0           | $-\infty$   | 0           |
| $\cot \theta$                 | $\infty$  | 0          | $\infty$    | 0           | $\infty$    |
| $\sec \theta$                 | 1         | $\infty$   | -1          | $\infty$    | 1           |
| $\operatorname{cosec} \theta$ | $\infty$  | 1          | $\infty$    | -1          | $\infty$    |

**Exercise 9.3****Q.1** Verify the following.

$$(i) \quad \sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ \\ = \sin 30^\circ$$

**Solution:**

L.H.S

$$= \sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$$

$$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{3}{4} - \frac{1}{4} = \frac{2}{4}$$

$$= \frac{1}{2} = \sin 30^\circ = \text{R.H.S.}$$

$$(ii) \quad \sin^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{3} + \tan^2 \frac{\pi}{4} = 2$$

**Solution:**

$$\text{L.H.S.} = \sin^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{3} + \tan^2 \frac{\pi}{4}$$

$$= \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2$$

$$= \frac{1}{4} + \frac{3}{4} + 1$$

$$= \frac{1+3+4}{4}$$

$$= \frac{8}{4}$$

$$= 2 = \text{R.H.S.}$$

$$(iii) \quad 2\sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ = \frac{3}{\sqrt{2}}$$

**Solution:**

$$\text{L.H.S.} = 2\sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ$$

$$= 2\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{2}(\sqrt{2})$$

$$= \frac{2}{\sqrt{2}} + \frac{\sqrt{2}}{2}$$

$$= \sqrt{2} + \frac{1}{\sqrt{2}}$$

$$= \frac{2+1}{\sqrt{2}}$$

$$= \frac{3}{\sqrt{2}} = \text{R.H.S.}$$

$$(iv) \quad \sin^2 \frac{\pi}{6} : \sin^2 \frac{\pi}{4} : \sin^2 \frac{\pi}{3} : \sin^2 \frac{\pi}{2} \\ = 1 : 2 : 3 : 4$$

**Solution:**

$$\text{L.H.S.} = \sin^2 \frac{\pi}{6} : \sin^2 \frac{\pi}{4} : \sin^2 \frac{\pi}{3} : \sin^2 \frac{\pi}{2}$$

$$= \left(\frac{1}{2}\right)^2 : \left(\frac{1}{\sqrt{2}}\right)^2 : \left(\frac{\sqrt{3}}{2}\right)^2 : (1)^2$$

$$= \frac{1}{4} : \frac{1}{2} : \frac{3}{4} : 1$$

Multiplying by 4, we get

$$= 1 : 2 : 3 : 4 = \text{R.H.S.}$$

**Q.2** Evaluate the following

$$(i) \quad \frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{3} \tan \frac{\pi}{6}}$$

**Solution:**

$$\frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{3} \tan \frac{\pi}{6}} = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + (\sqrt{3})\left(\frac{1}{\sqrt{3}}\right)}$$

$$= \frac{3-1}{1+1}$$

$$= \frac{2}{2}$$

$$= \frac{1}{1}$$

$$= 1$$



$$(ii) \quad \frac{1 - \tan^2 \frac{\pi}{3}}{1 + \tan^2 \frac{\pi}{3}}$$

**Solution:**

$$\frac{1 - \tan^2 \frac{\pi}{3}}{1 + \tan^2 \frac{\pi}{3}} = \frac{1 - (\sqrt{3})^2}{1 + (\sqrt{3})^2}$$

$$\begin{aligned} &= \frac{1-3}{1+3} \\ &= \frac{-2}{4} \\ &= \frac{-1}{2} \end{aligned}$$

**Q.3** Verify the following when  $\theta = 30^\circ, 45^\circ$

(i)  $\sin 2\theta = 2\sin\theta\cos\theta$

(a) When  $\theta = 30^\circ$

$$\text{L.H.S.} = \sin 2\theta$$

$$= \sin 2(30^\circ)$$

$$= \sin 60^\circ$$

$$= \frac{\sqrt{3}}{2}$$

$$\text{R.H.S.} = 2\sin\theta\cos\theta$$

$$= 2\sin 30^\circ\cos 30^\circ$$

$$= 2\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{3}}{2}$$

(b) When  $\theta = 45^\circ$

$$\text{L.H.S.} = \sin 2\theta$$

$$= \sin 2(45^\circ)$$

$$= \sin 90^\circ$$

$$= 1$$

$$\text{R.H.S.} = 2\sin\theta\cos\theta$$

$$= 2\sin 45^\circ\cos 45^\circ$$

$$= 2\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$= 1$$

(ii)  $\cos 2\theta = \cos^2\theta - \sin^2\theta$

(a) When  $\theta = 30^\circ$

$$\text{L.H.S.} = \cos 2\theta$$

$$= \cos 2(30^\circ)$$

$$= \cos 60^\circ$$

$$= \frac{1}{2}$$

$$\text{R.H.S.} = \cos^2\theta - \sin^2\theta$$

$$= \cos^2 30^\circ - \sin^2 30^\circ$$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{3-1}{4}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

(b) When  $\theta = 45^\circ$ 

$$\begin{aligned}\text{L.H.S.} &= \cos 2\theta \\ &= \cos 2(45^\circ) \\ &= \cos 90^\circ \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{R.H.S.} &= \cos^2 \theta - \sin^2 \theta \\ &= \cos^2 45^\circ - \sin^2 45^\circ \\ &= \left(\frac{1}{\sqrt{2}}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2 \\ &= \frac{1}{2} - \frac{1}{2} \\ &= 0\end{aligned}$$

(iii)  $\cos 2\theta = 2\cos^2 \theta - 1$ (a) When  $\theta = 30^\circ$ 

$$\begin{aligned}\text{L.H.S.} &= \cos 2\theta \\ &= \cos 2(30^\circ) \\ &= \cos 60^\circ \\ &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\text{R.H.S.} &= 2\cos^2 \theta - 1 \\ &= 2(\cos 30^\circ)^2 - 1 \\ &= 2\left(\frac{\sqrt{3}}{2}\right)^2 - 1 \\ &= 2\left(\frac{3}{4}\right) - 1 \\ &= \frac{3}{2} - 1 \\ &= \frac{3-2}{2} \\ &= \frac{1}{2}\end{aligned}$$

(b) When  $\theta = 45^\circ$ 

$$\begin{aligned}\text{L.H.S.} &= \cos 2\theta \\ &= \cos 2(45^\circ) \\ &= \cos 90^\circ \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{R.H.S.} &= 2\cos^2 \theta - 1 \\ &= 2\cos^2 45^\circ - 1 \\ &= 2\left(\frac{1}{2}\right)^2 - 1 \\ &= 0\end{aligned}$$

(iv)  $\cos 2\theta = 1 - 2\sin^2 \theta$ (a) When  $\theta = 30^\circ$ 

$$\begin{aligned}\text{L.H.S.} &= \cos 2\theta \\ &= \cos 2(30^\circ) \\ &= \cos 60^\circ \\ &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\text{R.H.S.} &= 1 - 2\sin^2 \theta \\ &= 1 - 2\sin^2 30^\circ \\ &= 1 - 2\left(\frac{1}{2}\right)^2 = 1 - \frac{1}{2} \\ &= \frac{1}{2}\end{aligned}$$

(b) When  $\theta = 45^\circ$ 

$$\text{L.H.S.} = \cos 2\theta$$

$$= \cos 2(45^\circ)$$

$$= \cos 90^\circ$$

$$= 0$$

$$\text{R.H.S.} = 1 - 2 \sin^2 \theta$$

$$= 1 - 2 \sin^2 45^\circ$$

$$= 1 - 2 \left( \frac{1}{2} \right)$$

$$= 0$$

$$(v) \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

(a) When  $\theta = 30^\circ$ 

$$\text{L.H.S.} = \tan 2\theta$$

$$= \tan 2(30^\circ)$$

$$= \tan 60^\circ$$

$$= \sqrt{3}$$

$$\text{R.H.S.} = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$$

$$= \frac{2 \left( \frac{1}{\sqrt{3}} \right)}{1 - \frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}}$$

$$= \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \frac{3}{\sqrt{3}}$$

$$= \sqrt{3}$$

(b) When  $\theta = 45^\circ$ 

$$\text{L.H.S.} = \tan 2\theta$$

$$= \tan 2(45^\circ)$$

$$= \tan 90^\circ$$

$$= \text{undefined}$$

$$\text{R.H.S.} = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \frac{2 \tan 45^\circ}{1 - \tan^2 (45^\circ)}$$

$$= \frac{2(1)}{1-1}$$

$$= \text{undefined}$$

**Q.4 Find the values of the trigonometric functions of the following quadrantal angles.**

**(i)  $-\pi$**

**Solution:**

$$-\pi = -\pi + 2\pi = \pi$$

So the values of trigonometric functions at  $-\pi$  are same as that of  $\pi$

$$\begin{array}{l|l} \sin(-\pi) = \sin \pi = 0 & \operatorname{cosec}(-\pi) = \operatorname{cosec} \pi = \text{undefined} \\ \cos(-\pi) = \cos \pi = -1 & \sec(-\pi) = \sec \pi = -1 \\ \tan(-\pi) = \tan \pi = 0 & \cot(-\pi) = \cot \pi = \text{undefined} \end{array}$$

**(ii)  $-3\pi$**

**Solution:**

$$-3\pi = -3\pi + 2(2\pi) = -3\pi + 4\pi = \pi$$

So the values of trigonometric function at  $-3\pi$  are same as that of  $\pi$ .

$$\begin{array}{l|l} \sin \pi = 0 & \operatorname{cosec} \pi = \text{undefined} \\ \cos \pi = -1 & \sec \pi = -1 \\ \tan \pi = 0 & \cot \pi = \text{undefined} \end{array}$$

**(iii)  $\frac{5\pi}{2}$**

**Solution:**

$$\begin{aligned} \frac{5\pi}{2} &= \frac{5\pi}{2} + (-1)(2\pi) = \frac{5\pi}{2} - 2\pi \\ &= \frac{5\pi - 4\pi}{2} = \frac{\pi}{2} \end{aligned}$$

So the values of trigonometric function at  $\frac{5\pi}{2}$  are same as that of  $\frac{\pi}{2}$ .

$$\begin{array}{l|l} \sin \frac{\pi}{2} = 1 & \operatorname{cosec} \frac{\pi}{2} = 1 \\ \cos \frac{\pi}{2} = 0 & \sec \frac{\pi}{2} = \infty \\ \tan \frac{\pi}{2} = \infty & \cot \frac{\pi}{2} = 0 \end{array}$$

(iv)  $-\frac{9\pi}{2}$

**Solution:**

$$-\frac{9\pi}{2} = -\frac{9\pi}{2} + 3(2\pi) = -\frac{9\pi}{2} + 6\pi$$

$$= \frac{-9\pi + 12\pi}{2} = \frac{3\pi}{2}$$

So the values of trigonometric function at  $-\frac{9\pi}{2}$  are same as that of  $\frac{3\pi}{2}$ .

|                                |  |                                            |
|--------------------------------|--|--------------------------------------------|
| $\sin \frac{3\pi}{2} = -1$     |  | $\operatorname{cosec} \frac{3\pi}{2} = -1$ |
| $\cos \frac{3\pi}{2} = 0$      |  | $\sec \frac{3\pi}{2} = \infty$             |
| $\tan \frac{3\pi}{2} = \infty$ |  | $\operatorname{cosec} \frac{3\pi}{2} = -1$ |

(v)  $-15\pi$

**Solution:**

$$-15\pi = -15\pi + 8(2\pi) = -15\pi + 16\pi = \pi$$

So the values of trigonometric function at  $-15\pi$  are same as that of  $\pi$ .

|                 |  |                                     |
|-----------------|--|-------------------------------------|
| $\sin \pi = 0$  |  | $\operatorname{cosec} \pi = \infty$ |
| $\cos \pi = -1$ |  | $\sec \pi = -1$                     |
| $\tan \pi = 0$  |  | $\cot \pi = \infty$                 |

(vi)  $1530^\circ$

**Solution:**

$$1530^\circ = 1530^\circ + (-4)(360^\circ) = 1530^\circ - 1440^\circ = 90^\circ$$

So the values of trigonometric function at  $1530^\circ$  are same as that of  $90^\circ$ .

|                          |  |                                     |
|--------------------------|--|-------------------------------------|
| $\sin 90^\circ = 1$      |  | $\operatorname{cosec} 90^\circ = 1$ |
| $\cos 90^\circ = 0$      |  | $\sec 90^\circ = \infty$            |
| $\tan 90^\circ = \infty$ |  | $\cot 90^\circ = 0$                 |

(vii)  $-2430^\circ$

**Solution:**

$$-2430^\circ = -2430^\circ + 7(360^\circ)$$

$$= -2430^\circ + 2520^\circ$$

$$= 90^\circ$$

So the values of trigonometric function at  $-2430^\circ$  are same as that of  $90^\circ$ .

|                          |  |                                     |
|--------------------------|--|-------------------------------------|
| $\sin 90^\circ = 1$      |  | $\operatorname{cosec} 90^\circ = 1$ |
| $\cos 90^\circ = 0$      |  | $\sec 90^\circ = \infty$            |
| $\tan 90^\circ = \infty$ |  | $\cot 90^\circ = 0$                 |

(viii)  $\frac{235\pi}{2}$

**Solution:**

$$\begin{aligned}\frac{235\pi}{2} &= \frac{235\pi}{2} + (-58)(2\pi) = \frac{235\pi}{2} - 116\pi \\ &= \frac{235\pi - 232\pi}{2} \\ &= \frac{3\pi}{2}\end{aligned}$$

So the values of trigonometric function at  $\frac{235\pi}{2}$  are same as that of  $\frac{3\pi}{2}$ .

$$\begin{array}{l|l}\sin \frac{3\pi}{2} = -1 & \operatorname{cosec} \frac{3\pi}{2} = -1 \\ \cos \frac{3\pi}{2} = 0 & \sec \frac{3\pi}{2} = \infty \\ \tan \frac{3\pi}{2} = \infty & \cot \frac{3\pi}{2} = 0\end{array}$$

(ix)  $\frac{407\pi}{2}$

**Solution:**

$$\begin{aligned}\frac{407\pi}{2} &= \frac{407\pi}{2} + (-101)(2\pi) = \frac{407\pi}{2} - 202\pi \\ &= \frac{407\pi - 404\pi}{2} \\ &= \frac{3\pi}{2}\end{aligned}$$

So the values of trigonometric function at  $\frac{407\pi}{2}$  are same as that of  $\frac{3\pi}{2}$ .

$$\begin{array}{l|l}\sin \frac{3\pi}{2} = -1 & \operatorname{cosec} \frac{3\pi}{2} = -1 \\ \cos \frac{3\pi}{2} = 0 & \sec \frac{3\pi}{2} = \infty \\ \tan \frac{3\pi}{2} = \infty & \cot \frac{3\pi}{2} = 0\end{array}$$

**Q.5 Find the values of trigonometric functions of the following angles.**

**(i)  $390^\circ$**

**Solution:**

$$390^\circ = 390^\circ + (-1)(360^\circ) = 390^\circ - 360^\circ = 30^\circ$$

So the values of trigonometric function at  $390^\circ$  are same as that of  $30^\circ$ .

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| $\sin 30^\circ = \frac{1}{2}$        | $\operatorname{cosec} 30^\circ = 2$  |
| $\cos 30^\circ = \frac{\sqrt{3}}{2}$ | $\sec 30^\circ = \frac{2}{\sqrt{3}}$ |
| $\tan 30^\circ = \frac{1}{\sqrt{3}}$ | $\cot 30^\circ = \sqrt{3}$           |

**(ii)  $-330^\circ$**

**Solution:**

$$-330^\circ = -330^\circ + 360^\circ = 30^\circ$$

So the values of trigonometric function at  $-330^\circ$  are same as that of  $30^\circ$ .

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| $\sin 30^\circ = \frac{1}{2}$        | $\operatorname{cosec} 30^\circ = 2$  |
| $\cos 30^\circ = \frac{\sqrt{3}}{2}$ | $\sec 30^\circ = \frac{2}{\sqrt{3}}$ |
| $\tan 30^\circ = \frac{1}{\sqrt{3}}$ | $\cot 30^\circ = \sqrt{3}$           |

**(iii)  $765^\circ$**

**Solution:**

$$765^\circ = 765^\circ + (-2)(360^\circ) = 765^\circ - 720^\circ = 45^\circ$$

So the values of trigonometric function at  $765^\circ$  are same as that of  $45^\circ$ .

|                                      |                                            |
|--------------------------------------|--------------------------------------------|
| $\sin 45^\circ = \frac{1}{\sqrt{2}}$ | $\operatorname{cosec} 45^\circ = \sqrt{2}$ |
| $\cos 45^\circ = \frac{1}{\sqrt{2}}$ | $\sec 45^\circ = \sqrt{2}$                 |
| $\tan 45^\circ = 1$                  | $\cot 45^\circ = 1$                        |

**(iv)  $-675^\circ$**

**Solution:**

$$-675^\circ = -675^\circ + 2(360^\circ) = -675^\circ + 720^\circ = 45^\circ$$

So the values of trigonometric function at  $-675^\circ$  are same as that of  $45^\circ$ .

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = 1$$

$$(v) \quad \frac{-17\pi}{3}$$

$$\operatorname{cosec} 45^\circ = \sqrt{2}$$

$$\sec 45^\circ = \sqrt{2}$$

$$\cot 45^\circ = 1$$

**Solution:**

$$-\frac{17\pi}{3} = \frac{-17\pi}{3} + 3(2\pi) = \frac{-17\pi}{3} + 6\pi = \frac{-17\pi + 18\pi}{3} = \frac{\pi}{3}$$

So the values of trigonometric function at  $-\frac{17\pi}{3}$  are same as that of  $\frac{\pi}{3}$ .

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\tan \frac{\pi}{3} = \sqrt{3}$$

$$\operatorname{cosec} \frac{\pi}{3} = \frac{2}{\sqrt{3}}$$

$$\sec \frac{\pi}{3} = 2$$

$$\cot \frac{\pi}{3} = \frac{1}{\sqrt{3}}$$

$$(vi) \quad \frac{13\pi}{3}$$

**Solution:**

$$\frac{13\pi}{3} = \frac{13\pi}{3} + (-2)(2\pi) = \frac{13\pi}{3} - 4\pi = \frac{13\pi - 12\pi}{3} = \frac{\pi}{3}$$

So the values of trigonometric function at  $\frac{13\pi}{3}$  are same as that of  $\frac{\pi}{3}$ .

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\tan \frac{\pi}{3} = \sqrt{3}$$

$$\operatorname{cosec} \frac{\pi}{3} = \frac{2}{\sqrt{3}}$$

$$\sec \frac{\pi}{3} = 2$$

$$\cot \frac{\pi}{3} = \frac{1}{\sqrt{3}}$$



(vii)  $\frac{25\pi}{6}$

**Solution:**

$$\frac{25\pi}{6} = \frac{25\pi}{6} + (-2)(2\pi) = \frac{25\pi}{6} - 4\pi = \frac{25\pi - 24\pi}{6} = \frac{\pi}{6}$$

So the values of trigonometric function at  $\frac{25\pi}{6}$  are same as that of  $\frac{\pi}{6}$ .

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| $\sin \frac{\pi}{6} = \frac{1}{2}$        | $\operatorname{cosec} \frac{\pi}{6} = 2$  |
| $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ | $\sec \frac{\pi}{6} = \frac{2}{\sqrt{3}}$ |
| $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$ | $\cot \frac{\pi}{6} = \sqrt{3}$           |

(viii)  $\frac{-71\pi}{6}$

**Solution:**

$$\frac{-71\pi}{6} = \frac{-71\pi}{6} + 6(2\pi) = \frac{-71\pi}{6} + 12\pi = \frac{-71\pi + 72\pi}{6} = \frac{\pi}{6}$$

So the values of trigonometric function at  $\frac{-71\pi}{6}$  are same as that of  $\frac{\pi}{6}$ .

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| $\sin \frac{\pi}{6} = \frac{1}{2}$        | $\operatorname{cosec} \frac{\pi}{6} = 2$  |
| $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ | $\sec \frac{\pi}{6} = \frac{2}{\sqrt{3}}$ |
| $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$ | $\cot \frac{\pi}{6} = \sqrt{3}$           |

(ix)  $-1035^\circ$

**Solution:**

$$-1035^\circ = -1035^\circ + 3(360^\circ) = -1035^\circ + 1080^\circ = 45^\circ$$

So the values of trigonometric function at  $-1035^\circ$  are same as that of  $45^\circ$ .

|                                      |                                            |
|--------------------------------------|--------------------------------------------|
| $\sin 45^\circ = \frac{1}{\sqrt{2}}$ | $\operatorname{cosec} 45^\circ = \sqrt{2}$ |
| $\cos 45^\circ = \frac{1}{\sqrt{2}}$ | $\sec 45^\circ = \sqrt{2}$                 |
| $\tan 45^\circ = 1$                  | $\cot 45^\circ = 1$                        |

**Domains of “Trigonometric Functions and Fundamental Identities”.**

We list the trigonometric functions and fundamental identities, learnt so far mentioning their domains as follows.

- (i)  $\sin \theta$  , for all  $\theta \in \mathbb{R}$
- (ii)  $\cos \theta$  , for all  $\theta \in \mathbb{R}$
- (iii)  $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq n\pi$   $n \in \mathbb{Z}$
- (iv)  $\sec \theta = \frac{1}{\cos \theta}$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq (2n+1)\frac{\pi}{2}$   $n \in \mathbb{Z}$
- (v)  $\tan \theta = \frac{\sin \theta}{\cos \theta}$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq (2n+1)\frac{\pi}{2}$   $n \in \mathbb{Z}$
- (vi)  $\cot \theta = \frac{\cos \theta}{\sin \theta}$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq n\pi$   $n \in \mathbb{Z}$
- (vii)  $\sin^2 \theta + \cos^2 \theta = 1$  , for all  $\theta \in \mathbb{R}$
- (viii)  $1 + \tan^2 \theta = \sec^2 \theta$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq (2n+1)\frac{\pi}{2}$   $n \in \mathbb{Z}$
- (ix)  $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$  , for all  $\theta \in \mathbb{R}$  but  $\theta \neq n\pi$   $n \in \mathbb{Z}$

**Exercise 9.4**

Prove the following identities, state the domain of  $\theta$  in each case.

**Q.1**  $\tan\theta + \cot\theta = \operatorname{cosec}\theta \sec\theta$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \tan\theta + \cot\theta \\ &= \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin^2\theta + \cos^2\theta}{\cos\theta \sin\theta} \\ &= \frac{1}{\cos\theta} \cdot \frac{1}{\sin\theta} \\ &= \operatorname{cosec}\theta \cdot \sec\theta = \text{R.H.S.} \end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$

**Q.2**  $\sec\theta \operatorname{cosec}\theta \sin\theta \cos\theta = 1$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \sec\theta \operatorname{cosec}\theta \sin\theta \cos\theta \\ &= \frac{1}{\sin\theta} \cdot \frac{1}{\cos\theta} \cdot \sin\theta \cos\theta \\ &= 1 = \text{R.H.S.} \end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$

**Q.3**  $\cos\theta + \tan\theta \sin\theta = \sec\theta$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \cos\theta + \tan\theta \cdot \sin\theta \\ &= \cos\theta + \frac{\sin\theta}{\cos\theta} \cdot \sin\theta \\ &= \frac{\cos^2\theta + \sin^2\theta}{\cos\theta} \\ &= \frac{1}{\cos\theta} \\ &= \sec\theta = \text{R.H.S.} \end{aligned}$$

Domain of

$\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

**Q.4**  $\operatorname{cosec}\theta + \tan\theta \sec\theta = \operatorname{cosec}\theta \sec^2\theta$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \operatorname{cosec}\theta + \tan\theta \sec\theta \\ &= \frac{1}{\sin\theta} + \frac{\sin\theta}{\cos\theta} \cdot \frac{1}{\cos\theta} \\ &= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta \cos^2\theta} \\ &= \frac{1}{\sin\theta} \cdot \frac{1}{\cos^2\theta} \\ &= \operatorname{cosec}\theta \cdot \sec^2\theta = \text{R.H.S.} \end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$

**Q.5**  $\sec^2\theta - \operatorname{cosec}^2\theta = \tan^2\theta - \cot^2\theta$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \sec^2\theta - \operatorname{cosec}^2\theta \\ &= 1 + \tan^2\theta - (1 + \cot^2\theta) \\ &(\because \sec^2\theta = 1 + \tan^2\theta \text{ and } \operatorname{cosec}^2\theta = 1 + \cot^2\theta) \\ &= 1 + \tan^2\theta - 1 - \cot^2\theta \\ &= \tan^2\theta - \cot^2\theta \\ &= \text{R.H.S.} \end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$

**Q.6**  $\cot^2\theta - \cos^2\theta = \cot^2\theta \cdot \cos^2\theta$

**Solution:**

$$\begin{aligned} \text{L.H.S.} &= \cot^2\theta - \cos^2\theta \\ &= \frac{\cos^2\theta}{\sin^2\theta} - \cos^2\theta \\ &= \cos^2\theta \left( \frac{1}{\sin^2\theta} - 1 \right) \\ &= \cos^2\theta \left( \frac{1 - \sin^2\theta}{\sin^2\theta} \right) \end{aligned}$$

$$= \left( \frac{\cos^2 \theta}{\sin^2 \theta} \right) \cdot (\cos^2 \theta)$$

$$= \cot^2 \theta \cdot \cos^2 \theta = \text{R.H.S.}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq n\pi, n \in \mathbb{Z}$

**Q.7**  $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$

**Solution:**

$$\text{L.H.S.} = (\sec \theta + \tan \theta)(\sec \theta - \tan \theta)$$

$$= \sec^2 \theta - \tan^2 \theta$$

$$= \sec^2 \theta - (\sec^2 \theta - 1)$$

$$= \sec^2 \theta - \sec^2 \theta + 1$$

$$= 1 = \text{R.H.S.}$$

Domain of  $\theta$

:  $\theta \in \mathbb{R}$ , but  $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

**Q.8**  $2\cos^2 \theta - 1 = 1 - 2\sin^2 \theta$

**Solution:**

$$\text{L.H.S.} = 2\cos^2 \theta - 1$$

$$= 2(1 - \sin^2 \theta) - 1$$

$$= 2 - 2\sin^2 \theta - 1$$

$$= 1 - 2\sin^2 \theta$$

$$= \text{R.H.S.}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$

**Q.9**  $\cos^2 \theta - \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

**Solution:**

$$\text{R.H.S.} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$= \frac{1 - \tan^2 \theta}{\sec^2 \theta}$$

$$= \cos^2 \theta \left( 1 - \frac{\sin^2 \theta}{\cos^2 \theta} \right)$$

$$= \cos^2 \theta \left( \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \right)$$

$$= \cos^2 \theta - \sin^2 \theta$$

$$= \text{L.H.S.}$$

Domain of  $\theta$  :

$\theta \in \mathbb{R}$ , but  $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

**Q.10**  $\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \frac{\cot \theta - 1}{\cot \theta + 1}$

**Solution:**

$$\text{R.H.S.} = \frac{\cot \theta - 1}{\cot \theta + 1}$$

$$= \frac{\frac{\cos \theta}{\sin \theta} - 1}{\frac{\cos \theta}{\sin \theta} + 1}$$

$$= \frac{\frac{\cos \theta - \sin \theta}{\sin \theta}}{\frac{\cos \theta + \sin \theta}{\sin \theta}}$$

$$= \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$$

$$= \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \text{L.H.S.}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq n\pi, n \in \mathbb{Z}$

**Q.11**  $\frac{\sin \theta}{1 + \cos \theta} + \cot \theta = \operatorname{cosec} \theta$

**Solution:**

$$\text{L.H.S.} = \frac{\sin \theta}{1 + \cos \theta} + \cot \theta$$

$$= \frac{\sin \theta}{1 + \cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos \theta(1 + \cos \theta)}{\sin \theta(1 + \cos \theta)}$$

$$= \frac{\sin^2 \theta + \cos \theta + \cos^2 \theta}{\sin \theta(1 + \cos \theta)}$$

$$\begin{aligned}
 &= \frac{1 + \cos \theta}{\sin \theta (1 + \cos \theta)} \\
 &= \frac{1}{\sin \theta} \\
 &= \operatorname{cosec} \theta = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$ ;  $\theta \in \mathbb{R}$ , but  $\theta \neq n\pi, n \in \mathbb{Z}$

**Q.12**  $\frac{\cot^2 \theta - 1}{1 + \cot^2 \theta} = 2\cos^2 \theta - 1$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= \frac{\cot^2 \theta - 1}{1 + \cot^2 \theta} \\
 &= \frac{\cot^2 \theta - 1}{\operatorname{cosec}^2 \theta} \\
 &= \sin^2 \theta \left( \frac{\cos^2 \theta}{\sin^2 \theta} - 1 \right) \\
 &= \sin^2 \theta \left( \frac{\cos^2 \theta - \sin^2 \theta}{\sin^2 \theta} \right) \\
 &= \cos^2 \theta - \sin^2 \theta \\
 &= \cos^2 \theta - (1 - \cos^2 \theta) \\
 &= \cos^2 \theta - 1 + \cos^2 \theta \\
 &= 2\cos^2 \theta - 1 = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$ :  $\theta \in \mathbb{R}$ , but  $\theta \neq n\pi, n \in \mathbb{Z}$

**Q.13**  $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2$

**Solution:**

$$\begin{aligned}
 \text{R.H.S.} &= (\operatorname{cosec} \theta + \cot \theta)^2 \\
 &= \left( \frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} \right)^2 \\
 &= \left( \frac{1 + \cos \theta}{\sin \theta} \right)^2 \\
 &= \frac{(1 + \cos \theta)^2}{\sin^2 \theta}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta} \\
 &= \frac{(1 + \cos \theta) \cancel{(1 + \cos \theta)}}{(1 - \cos \theta) \cancel{(1 + \cos \theta)}} \\
 &= \frac{1 + \cos \theta}{1 - \cos \theta} = \text{L.H.S.}
 \end{aligned}$$

Domain of  $\theta$ :  $\theta \in \mathbb{R}$ , but  $\theta \neq n\pi, n \in \mathbb{Z}$

**Q.14**  $(\sec \theta - \tan \theta)^2 = \frac{1 - \sin \theta}{1 + \sin \theta}$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= (\sec \theta - \tan \theta)^2 \\
 &= \left( \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \right)^2 \\
 &= \left( \frac{1 - \sin \theta}{\cos \theta} \right)^2 \\
 &= \frac{(1 - \sin \theta)^2}{\cos^2 \theta} \\
 &= \frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta} \\
 &= \frac{\cancel{(1 - \sin \theta)} (1 - \sin \theta)}{\cancel{(1 - \sin \theta)} (1 + \sin \theta)} \\
 &= \frac{1 - \sin \theta}{1 + \sin \theta} = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$ :

$$\theta \in \mathbb{R}, \text{ but } \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

**Q.15**  $\frac{2\tan \theta}{1 + \tan^2 \theta} = 2\sin \theta \cos \theta$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= \frac{2 \tan \theta}{1 + \tan^2 \theta} \\
 &= \frac{2 \tan \theta}{\sec^2 \theta}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \sin \theta}{\cos \theta} \cdot \cos^2 \theta \\
 &= 2 \sin \theta \cos \theta \\
 &= \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$  :

$$\theta \in \mathbb{R}, \text{ but } \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

**Q.16**  $\frac{1 - \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= \frac{1 - \sin \theta}{\cos \theta} \\
 &= \frac{1 - \sin \theta}{\cos \theta} \cdot \frac{1 + \sin \theta}{1 + \sin \theta} \\
 &= \frac{(1 - \sin \theta)(1 + \sin \theta)}{\cos \theta(1 + \sin \theta)} \\
 &= \frac{1 - \sin^2 \theta}{\cos \theta(1 + \sin \theta)} \\
 &= \frac{\cos^2 \theta}{\cos \theta(1 + \sin \theta)} \\
 &= \frac{\cos \theta}{1 + \sin \theta} = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$  :

$$\theta \in \mathbb{R}, \text{ but } \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

**Q.17**  $(\tan \theta + \cot \theta)^2 = \sec^2 \theta \operatorname{cosec}^2 \theta$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= (\tan \theta + \cot \theta)^2 \\
 &= \left( \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)^2 \\
 &= \left( \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \right)^2 \\
 &= \left( \frac{1}{\cos \theta \sin \theta} \right)^2
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\cos^2 \theta} \cdot \frac{1}{\sin^2 \theta} \\
 &= \sec^2 \theta \operatorname{cosec}^2 \theta = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$

**Q.18**  $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \tan \theta + \sec \theta$

**Solution:**

$$\begin{aligned}
 \text{L.H.S.} &= \frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} \\
 &= \frac{\tan \theta + \sec \theta - (\sec^2 \theta - \tan^2 \theta)}{\tan \theta - \sec \theta + 1} \\
 &= \frac{(\tan \theta + \sec \theta) - (\sec \theta - \tan \theta)(\sec \theta + \tan \theta)}{\tan \theta - \sec \theta + 1} \\
 &= \frac{(\tan \theta + \sec \theta)[1 - (\sec \theta - \tan \theta)]}{\tan \theta - \sec \theta + 1} \\
 &= \frac{(\tan \theta + \sec \theta)[1 - \sec \theta + \tan \theta]}{\tan \theta - \sec \theta + 1} \\
 &= \frac{(\tan \theta + \sec \theta)(\tan \theta - \sec \theta + 1)}{\tan \theta - \sec \theta + 1} \\
 &= \tan \theta + \sec \theta = \text{R.H.S.}
 \end{aligned}$$

Domain of  $\theta$  :

$$\theta \in \mathbb{R}, \text{ but } \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

**Q.19**  $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = \frac{1}{\sin \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta}$

**Solution:**

We are to prove that

$$\begin{aligned}
 &\frac{1}{\operatorname{cosec} \theta - \cot \theta} + \frac{1}{\operatorname{cosec} \theta + \cot \theta} = \frac{2}{\sin \theta} \\
 \text{L.H.S.} &= \frac{1}{\operatorname{cosec} \theta - \cot \theta} + \frac{1}{\operatorname{cosec} \theta + \cot \theta}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\operatorname{cosec} \theta + \cot \theta + \operatorname{cosec} \theta - \cot \theta}{(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta + \cot \theta)} \\
&= \frac{2 \operatorname{cosec} \theta}{\operatorname{cosec}^2 \theta - \cot^2 \theta} \\
&= \frac{2 \operatorname{cosec} \theta}{1 + \cot^2 \theta - \cot^2 \theta} \\
&= \frac{2 \operatorname{cosec} \theta}{1} \\
&= \frac{2}{\sin \theta} = \text{R.H.S}
\end{aligned}$$

**Q.20**

$$\sin^3 \theta - \cos^3 \theta = (\sin \theta - \cos \theta)(1 + \sin \theta \cos \theta)$$

**Solution:**

$$\begin{aligned}
\text{L.H.S.} &= \sin^3 \theta - \cos^3 \theta \\
&= (\sin \theta)^3 - (\cos \theta)^3 \\
&= (\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta) \\
&= (\sin \theta - \cos \theta)(1 + \sin \theta \cos \theta) \\
&= \text{R.H.S.} \\
\text{Domain of } \theta : \theta &= \square
\end{aligned}$$

$$\text{Q.21} \quad \sin^6 \theta - \cos^6 \theta = (\sin^2 \theta - \cos^2 \theta)(1 - \sin^2 \theta \cos^2 \theta)$$

**Solution:**

$$\begin{aligned}
\text{L.H.S.} &= \sin^6 \theta - \cos^6 \theta \\
&= (\sin^2 \theta)^3 - (\cos^2 \theta)^3 \\
&= (\sin^2 \theta - \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta + \sin^2 \theta \cos^2 \theta) \\
&= (\sin^2 \theta - \cos^2 \theta) \left[ (\sin^2 \theta)^2 + (\cos^2 \theta)^2 + 2 \sin^2 \theta \cos^2 \theta - \sin^2 \theta \cos^2 \theta \right] \\
&= (\sin^2 \theta - \cos^2 \theta) \left[ (\sin^2 \theta + \cos^2 \theta)^2 - \sin^2 \theta \cos^2 \theta \right] \\
&= (\sin^2 \theta - \cos^2 \theta)(1 - \sin^2 \theta \cos^2 \theta) \\
&= \text{R.H.S.}
\end{aligned}$$

Domain of  $\theta : \theta \in \square$ 

$$\text{Q:22} \quad \sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$$

**Solution:**

$$\begin{aligned}
\text{L.H.S.} &= \sin^6 \theta + \cos^6 \theta \\
&= (\sin^2 \theta)^3 + (\cos^2 \theta)^3 \\
&= (\sin^2 \theta + \cos^2 \theta) [\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta] \\
&= (1) \left[ (\sin^2 \theta)^2 + (\cos^2 \theta)^2 + 2 \sin^2 \theta \cos^2 \theta - 3 \sin^2 \theta \cos^2 \theta \right] \\
&= (\sin^2 \theta + \cos^2 \theta)^2 - 3 \sin^2 \theta \cos^2 \theta \\
&= 1 - 3 \sin^2 \theta \cos^2 \theta = \text{R.H.S.}
\end{aligned}$$

Domain of  $\theta : \theta \in \square$

Q:23  $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} = 2\sec^2\theta$

**Solution:**

$$\begin{aligned}\text{L.H.S.} &= \frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} \\ &= \frac{1-\sin\theta+1+\sin\theta}{(1+\sin\theta)(1-\sin\theta)} \\ &= \frac{2}{1-\sin^2\theta} \\ &= \frac{2}{\cos^2\theta} \\ &= 2\sec^2\theta = \text{R.H.S.}\end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

Q:24  $\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} + \frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} = \frac{2}{1-2\sin^2\theta}$

**Solution:**

$$\begin{aligned}\text{L.H.S.} &= \frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} + \frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} \\ &= \frac{(\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2}{(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)} \\ &= \frac{\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta + \cos^2\theta + \sin^2\theta - 2\cos\theta\sin\theta}{\cos^2\theta - \sin^2\theta} \\ &= \frac{2(\cos^2\theta + \sin^2\theta)}{\cos^2\theta - \sin^2\theta} \\ &= \frac{2(\cos^2\theta + \sin^2\theta)}{1 - \sin^2\theta - \sin^2\theta} \\ &= \frac{2}{1 - \sin^2\theta - \sin^2\theta} \\ &= \frac{2}{1 - 2\sin^2\theta} \\ &= \text{R.H.S.}\end{aligned}$$

Domain of  $\theta$  :  $\theta \in \mathbb{R}$ , but  $\theta \neq (2n+1)\frac{\pi}{4}, n \in \mathbb{Z}$